

# Enhancing Multi-Modal Entity Alignment via Multi-Grained Decision Fusion

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## Abstract

Multi-modal entity alignment (MMEA) aims to identify equivalent entities across heterogeneous multi-modal knowledge graphs (MMKGs), which play a crucial role in organizing and integrating web knowledge from diverse modalities. Although prior studies have made progress by multi-modal features fusion, three inherent limitations remain unresolved. First, instance-level feature fusion is misaligned with the pair-wise task format of MMEA. Second, joint representations often overlook modality-specific characteristics, resulting in insufficient alignment. Third, most existing methods rely solely on global features of modality. This may lead to the misalignment of entities that are similar yet distinct. To address above issues, we propose *DMEA*, a new decision-fusion-based framework. Specifically, we first design a multi-modal knowledge encoding module to extract both global and local features for different modalities and then introduce a multi-grained alignment module, which consists of two components: intra-modal alignment and cross-modal alignment. The former computes alignment scores between the global and local features of entity pairs within the same modality, while the latter leverages the complementarity across modalities to compute cross-modal alignment scores. Each score is regarded as

an independent decision, and the final alignment judgment is made by integrating all decisions. Finally, we incorporate an intra-modal contrastive loss to obtain more discriminative embedding representations. *DMEA* achieves improvements of 13.2% and 14.8% in hit@1 over the state-of-the-art models on two benchmark datasets, FB15K-DB15K and FB15K-YAGO15K, respectively, validating the superiority of our framework. Our code is publicly available at <https://github.com/mikumifa/DMEA>.

## CCS Concepts

• Computing methodologies → Knowledge representation and reasoning.

## Keywords

Multi-Modal Entity Alignment, Decision Fusion, Multi-Grained Alignment, Contrastive Learning

## ACM Reference Format:

Yu Xing, Qizhuo Xie, You Lv, Ziyang Zhou, Qianzi Hou, Qing Gu, Bin Chong, and Tieke He. 2026. Enhancing Multi-Modal Entity Alignment via Multi-Grained Decision Fusion. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3774904.3792452>

## 1 Introduction

Multi-Modal Knowledge Graphs (MMKGs) [42] organize real-world knowledge in a structured manner across various modalities such as vision, graph structure, and relational attribute text. Their development has significantly advanced a wide range of web-based downstream applications, including multi-modal recommendation systems [11, 32, 38], question-answering systems [33, 36, 43], and

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WWW '26, Dubai, United Arab Emirates

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ACM ISBN 979-8-4007-2307-0/2026/04

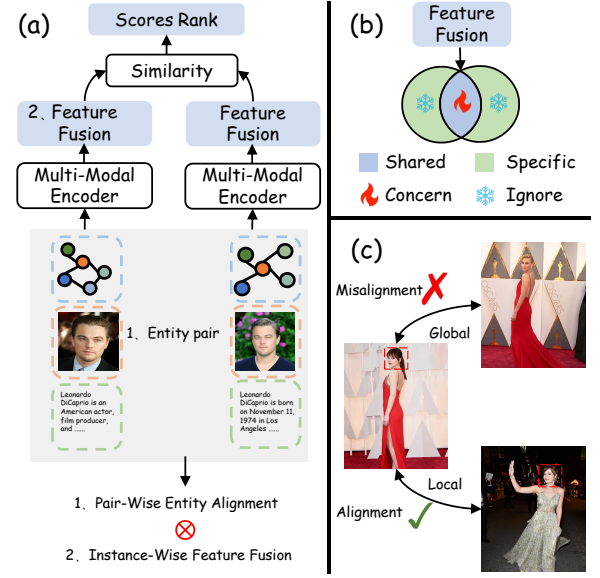
<https://doi.org/10.1145/3774904.3792452>

cross-modal retrieval [5, 17, 23]. However, since different MMKGs are constructed for different purposes across diverse knowledge sources, the same concept may be represented in multiple ways. Consequently, to build a comprehensive and complete MMKG, the integration of MMKGs from heterogeneous data sources has become an urgent requirement. As a result, Multi-Modal Entity Alignment (MMEA) has emerged as a critical task, which aims to align entities referring to the same real-world concept across two MMKGs using multi-modal information.

Recent methods for MMEA all follow a common framework (shown in Figure 1(a)): they first fused multi-modal information of entity to obtain a joint representation, and then computing the entity similarity for alignment. In this framework, the way of feature fusion has become the central focus. For instance, Chen et al. [4] perform a simple concatenation for each modality. Subsequent studies further explore weighted concatenation [18, 19], weighted summation [15], attention mechanisms [7], as well as structure- or vision-guided encoding strategies [3] to enhance the entity joint representation. This strategy belongs to feature-level fusion in multi-modal representation learning [41]. Although these approaches have achieved remarkable progress, this mechanism still suffers from three inherent limitations.

First, feature-level fusion is misaligned with the task format. Since MMEA is a pair-wise entity alignment task where entities from two KGs independently possess multi-modal information, performing instance-wise fusion only within a single entity is insufficient for cross-KG alignment. Second, joint representations suffer from alignment deficiencies. As shown in Figure 1(b), during feature fusion, modality-specific characteristics are frequently overlooked [8, 13], rendering the joint representation insufficient for supporting accurate entity alignment. These observations indicate that relying on a joint vector obtained through feature fusion is insufficient for effective entity alignment. Finally, models struggle to capture fine-grained alignment. Most existing methods focus solely on global representations of each modality [14]. This risks confusing entities that are visually or semantically similar yet distinct. As illustrated in Figure 1(c), highly similar scenes (e.g., red-carpet photos) may cause misalignment, whereas local details such as facial features are crucial for correct alignment.

To overcome the above limitations, we propose *DMEA*, a new MMEA framework based on decision fusion. Unlike traditional methods that rely solely on joint representations obtained through feature fusion to compute entity similarity, our approach models multi-granularity alignment relationships based on both global and local features, and performs entity alignment through decision fusion. This design fully exploits the multimodal alignment relationships between entity pairs, leading to breakthrough improvements on the MMEA task. Figure 2 illustrates our decision-fusion-based framework. Concretely, *DMEA* consists of three main modules. (1) Multi-modal knowledge encoding. This module employs dedicated encoders to extract both global and local features for each modality, enabling the model to capture holistic semantics while preserving fine-grained signals, thereby preparing for multi-grained alignment. (2) Multi-grained alignment module. This module explicitly models diverse alignment types to enhance the reliability of the alignment. It comprises two components: intra-modal alignment and cross-modal alignment. The former computes similarity scores



**Figure 1: (a) Common framework of current MMEA methods; (b) Feature fusion tends to overlook modality-specific information, thereby limiting model performance; (c) Relying solely on global feature may lead to incorrect alignments, while local feature provide the fine-grained details.**

between the global and local features of entity pairs within the same modality, while the latter leverages multi-modal complementarity to derive additional alignment scores across modalities. Each score is regarded as an independent decision, and entity alignment is inferred by fusing these decisions. (3) Intra-modal contrastive loss. Finally, we incorporate an intra-modal contrastive loss to learn more discriminative embeddings within each modality. Extensive experiments conducted on two benchmark MMEA datasets, FB15K-DB15K and FB15K-YAGO15K, demonstrate the superior performance and effectiveness of *DMEA*. The main contributions of our work can be summarized as follows:

- We propose *DMEA*, a decision-fusion based MMEA framework that integrates multi-grained alignment modules, including intra-modal and cross-modal scores, to provide a more reliable and holistic basis for entity alignment.
- Global-local multi-modal encoding is designed to capture both comprehensive semantics and fine-grained details, enabling the model to capture more fine-grained alignment relationships between entity pairs.
- Experiments show that *DMEA* outperforms the state-of-the-art models by 13.2% and 14.8% in Hits@1 on the FB15K-DB15K and FB15K-YAGO15K benchmark datasets, respectively. This significant breakthrough verifies the remarkable superiority of our strategy on the MMEA task.

## 2 Related Work

### 2.1 Multi-Modal Entity Alignment

Multi-Modal Entity Alignment aims to identify same entities across different knowledge graphs by introducing multiple modalities.

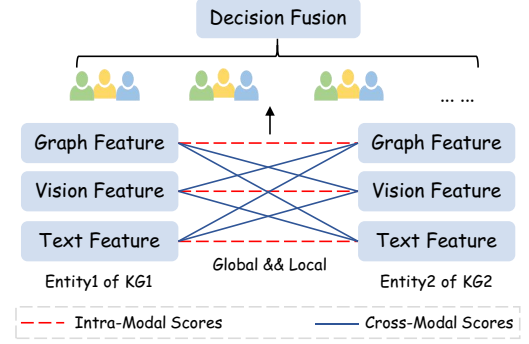
Existing approaches primarily focus on enhancing modality encoding and exploring multi-modal fusion strategies. MMEA [2] was one of the earliest attempts, where different modalities are simply projected into a shared embedding space. EVA [19] introduced a learnable weighting mechanism for different modality to dynamically adjust modality contributions. MSNEA [3] further exploited visual signals as auxiliary guidance for learning representations of relations and attributes. MEAformer [6] advanced this line by employing a multi-modal transformer that adaptively predicts modality-specific importance at the entity level for feature integration. To alleviate modality gaps, MCLEA [18] incorporated contrastive learning across modalities. PCMEA [29] proposed a pseudo-label calibration mechanism with momentum-based contrastive learning to leverage unlabeled samples more effectively. SGMEA [7] proposes a structure-guided MMEA framework that leverages the structural information of KGs to enhance visual and attribute modalities, thereby improving cross-modal interaction. Most recently, RICEA [16] investigated interaction-aware weighting schemes and refined them through uncertainty estimation to dynamically adjust modality contributions. Despite these studies have achieved remarkable progress, they primarily focus on intra-entity feature fusion while overlooking its inherent limitations in the MMEA task. Furthermore, existing methods emphasize only the global representations of each modality and neglect local alignment relationships, which may impair model performance. In this paper, we incorporate both global and local features and, inspired by the concept of decision fusion, model multi-granularity alignment relationships between entities.

## 2.2 Multi-Modal Fusion

Multi-modal fusion [10, 41] integrates multi-modal information (e.g., text, image, audio) to improve task performance. According to the levels of fusion, there are three strategies: data fusion (early fusion), feature fusion (intermediate fusion) and decision fusion (late fusion). Data fusion [9, 20, 37] directly integrates the raw data from each modality before sending it to a single encoder, which is simple but limited to homogeneous data types, requiring additional preprocessing for heterogeneous data. Instead of fusing raw data, feature fusion [26, 30, 40] fuses features from different encoders and then multi-modal representations are mapped into the joint space. Therefore, feature fusion is able to capture interactions among modalities. Decision fusion [1, 21, 35] is another common strategy, which integrates the outputs from individual modality-specific decoder at the decision stage. Previous MMEA methods primarily focused on feature fusion. However, due to the independent multi-modal information on both sides of the entity pairs in MMEA, feature fusion is not well-suited to the task. In this paper, we explore the potential of decision-level fusion for the MMEA task and propose a fine-grained, multi-grained alignment framework.

## 3 Method

In this section, we first introduce the MMEA task definition, and then introduce the proposed framework, including: multi-modal knowledge encoding, multi-grained alignment module and intra-modal contrastive module.



**Figure 2: A multi-grained alignment framework for MMEA based on decision fusion. Unlike traditional methods that rely solely on the final joint representation to compute similarity for entity alignment, we model both intra-modal and cross-modal multi-granularity alignments based on global and local features.**

### 3.1 Preliminary

**3.1.1 Multi-Modal Knowledge Graph.** A multi-modal knowledge graph (MMKG) extends traditional knowledge graphs by incorporating not only relational and attribute information but also visual content. Formally, it can be defined as  $G = \{E, R, I, A, V, T_r, T_a\}$ , where  $E, R, I, A$ , and  $V$  denote the sets of entities, relations, images, attributes, and attribute values, respectively. The set of relational triples is given by  $T_r = \{(e_h, r, e_t) | e, e' \in E, r \in R\}$  while the set of attribute triples is defined as  $T_a = \{(e, a, v) | e \in E, a \in A, v \in V\}$ .

**3.1.2 Multi-Modal Entity Alignment.** The objective of multi-modal entity alignment (MMEA) is to identify pairs of entities across two MMKGs that refer to the same real-world concept by exploiting information from multiple modalities. Formally, given two MMKGs:  $\mathcal{G}_1 = \{E_1, R_1, I_1, A_1, V_1, T_r^1, T_a^1\}$  and  $\mathcal{G}_2 = \{E_2, R_2, I_2, A_2, V_2, T_r^2, T_a^2\}$ , the alignment task aims to determine the set of equivalent entity pairs, denoted as:

$$\mathcal{S} = \{(e_1, e_2) \mid e_1 \equiv e_2, e_1 \in E_1, e_2 \in E_2\}. \quad (1)$$

In practice, only a subset of  $\mathcal{S}$  is typically provided as seed alignments for supervised training, and the model is expected to align the remaining entity pairs.

### 3.2 Multi-Modal Knowledge Encoding

Prior studies focus on global features, which may lead to confusion among similar entities. In this module, we design an encoder that captures both global and local features for each modalities, thereby enabling more precise alignment.

**3.2.1 Graph Modal Embedding.** For graph structure, we employ Graph Attention Network (GAT) [28] as the encoder. GAT adaptively assigns different attention weights to the entity neighbors, thereby capturing the complex interactions. Specifically, for a given entity, its vector is randomly initialized as  $h^g$ , and then updated through weighted aggregation of its  $n$ -hop neighbors as follows:

$$h_{l+1}^g = \text{GAT}(h_l^g, a) \in d_g, \quad (2)$$

where  $l$  denotes the number of propagation hops,  $a$  represents the adjacency matrix and  $d_g$  is the dimension of graph structural feature. We take the final aggregated representation as the global graph feature  $h_G^g$ , while the intermediate representations  $\{h_1^g, h_2^g, \dots, h_n^g\}$  from each hop are regarded as local features  $h_L^g$ .

**3.2.2 Textual and Visual Modal Embedding.** For the textual and visual modality, to mitigate the heterogeneity between modalities, we adopt the CLIP [24] encoder, which aligns the two modalities within a shared semantic space through contrastive learning. Specifically, we construct the textual description of each entity in the form of a natural language sentence, where the entity name and its attributes are concatenated sequentially. The description is formulated as: “The  $e_a$  of  $e_N$  is  $v_a$ ”. All attributes are concatenated in this manner to form a complete entity description. Then, we prepend the special token [CLS] and feed the sequence into CLIP to produce the hidden states of all tokens:

$$\{h_{[\text{CLS}]}^t, h_1^t, h_2^t, \dots, h_n^t\} \in \mathbb{R}^{(n+1) \times d_t}, \quad (3)$$

where  $d_t$  is the dimension of textual features. Then, we take  $h_{[\text{CLS}]}^t$  as the textual global feature  $h_G^t$  and the rest embeddings as local textual features  $h_L^t$ .

For visual modality, given an entity image, we reshape it into a sequence of image patches and prepend an additional [CLS] token before the image patches to represent the visual global feature and feed the sequence into CLIP to produce the hidden embeddings:

$$\{h_{[\text{CLS}]}^v, h_1^v, h_2^v, \dots, h_m^v\} \in \mathbb{R}^{(m+1) \times d_v}.$$

We take  $h_{[\text{CLS}]}^v$  as the visual global feature  $h_G^v$  and the rest embeddings as local features  $h_L^v$ .

### 3.3 Multi-Grained Alignment Module

To mitigate the potential drawbacks of feature fusion, we draw inspiration from decision-level fusion (refer to section 2.2 for details). Specifically, we design two alignment modules: (1) the intra-modal alignment module, which includes coarse-grained global-to-global alignment and fine-grained local-to-local alignment; and (2) the cross-modal alignment module, which exploits the complementarity among modalities to uncover latent semantic knowledge. Each module takes the representations of an entity pair  $(e_1, e_2)$  as input and computes an alignment score as follows:

$$S_I = \sum_{m \in \{g, v, t\}} S_m(e_1, e_2) = \sum_{m \in \{g, v, t\}} (S_m^{G2G} + S_m^{L2L})/2, \quad (4)$$

$$S_C = \sum_{m \in \{gt, tg, tv, vt\}} S_m(e_1, e_2)/4, \quad (5)$$

$$S = S(e_1, e_2) = (S_I + S_C)/2. \quad (6)$$

Each score can be viewed as an individual alignment decision, and the final alignment confidence is inferred by averaging across all decisions. Next, we elaborate on them in detail one by one.

**3.3.1 Intra-modal Alignment sub-Module.** In this module, given an input entity pair  $(e_1, e_2)$ , we compute the alignment score within the same modality. The score consists of two components: (i) coarse-grained global-to-global alignment and (ii) fine-grained local-to-local alignment. Taking the graph structural modality as an example,

given the global and local features of  $e_1$  as  $h_G^{g,1}$  and  $h_L^{g,1}$ , and  $e_2$  as  $h_G^{g,2}$  and  $h_L^{g,2}$ . The computation is formulated as follows:

**Coarse-grained Global-to-Global Alignment.** First, we compute the global alignment score by applying a similarity function to global textual vectors  $h_G^{g,1}$  and  $h_G^{g,2}$ .

$$S_t^{G2G} = f(h_G^{g,1}, h_G^{g,2}). \quad (7)$$

**Fine-grained Local-to-Local Alignment.** Then, we introduce an attention mechanism [27] to capture the fine-grained alignment between the local features, as follows:

$$\begin{aligned} Q_g, K_g, V_g &= W_1(h_L^{g,1}), W_2(h_L^{g,2}), W_3(h_L^{g,2}), \in \mathbb{R}^{d_s} \\ h_{L2L}^t &= \text{AP}\left(\text{Softmax}\left(\frac{Q_g K_g^T}{\sqrt{d_s}}\right) V_g\right), \\ S_g^{L2L} &= f(h_G^{g,1}, h_{L2L}^g), \end{aligned} \quad (8)$$

where  $\{W_1, W_2, W_3\}$  denotes the projection matrix and  $d_s$  denotes the scaled dimension. The operator  $\text{AP}(\cdot)$  performs average pooling over each dimension. The attention-weighted local features  $h_{L2L}^g$  are constrained with the global features to compute the alignment score. Then, we average the global-to-global and local-to-local scores to obtain the textual intra-modal alignment score:

$$S_g = (S_g^{G2G} + S_g^{L2L})/2. \quad (9)$$

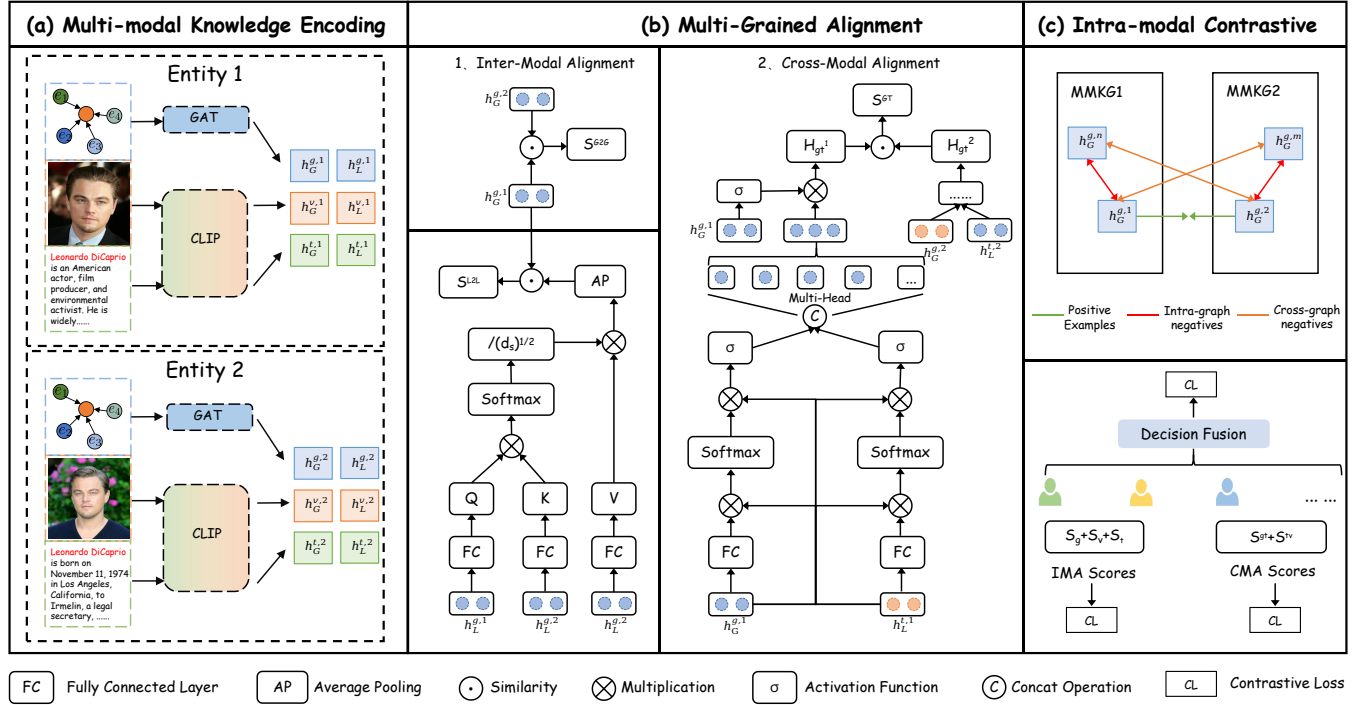
Similarly, for the textual and visual modality, we can obtain the alignment score  $S_t$  and  $S_v$ .

**3.3.2 Cross-modal Alignment Module.** A single modality may contain implicit information that can only be inferred through multi-modal analysis. Hence, we design a multi-head attention fusion mechanism that highlights complementary cues across modalities, thereby modeling cross-modal alignment more effectively.

Taking graph-to-text cross-modal alignment as an example and considering an entity pair  $(e_1, e_2)$ , for entity  $e_1$ , we first perform multi-modal fusion between its graph structural global features  $h_G^{g,1}$  and textual local features  $h_L^{t,1}$ . Both features are projected into a shared latent space via a learnable projection matrix, and then an attention mechanism is introduced to realize bidirectional cross-modal enhancement, resulting in local textual-aware global attention  $H_G^{g,1}$  and global graph-aware local attention  $H_L^{t,1}$ , respectively. Finally, we utilize the attention content to get an enhanced entity global graph feature  $H_{G'}^{g,1}$  and local textual features  $H_{L'}^{t,1}$ , as follows:

$$\begin{aligned} h_{G'}^{t,1}, h_{L'}^{v,1} &= W_4(h_G^{t,1}), W_5(h_L^{v,1}) \in \mathbb{R}^{d_c}, \\ H_{GL}^t, H_{LG}^t &= \text{Softmax}(h_{G'}^{t,1} \otimes h_{L'}^{v,1}), \text{Softmax}(h_{L'}^{v,1} \otimes h_{G'}^{t,1}), \\ h_{G'}^{t,1}, h_{L'}^{v,1} &= \sigma(H_{GL}^t \otimes h_{L'}^{v,1}), \sigma(H_{LG}^t \otimes h_{G'}^{t,1}), \end{aligned} \quad (10)$$

where  $d_c$  represents the scaled dimension size,  $\otimes$  is matrix multiplication. Then, given the  $H_{G'}^{g,1}$  and  $H_{L'}^{t,1}$ , we concatenate them and then introduce a multi-head attention mechanism to convert the concatenated vector  $h_{gt}^1$  into  $H_{gt}^1$ . Additionally, we apply a gating operation to fuse  $H_{gt}^1$  and the original global graph feature  $h_G^{g,1}$  to



**Figure 3: The overview of our framework, which first incorporates a multi-modal knowledge encoding module to extract both global and local feature of modality. Second, a multi-grained alignment module is designed, comprising intra-modal and cross-modal alignment. The final entity alignment is determined by jointly integrating the decisions from each part. Finally, we introduce a intra-modal contrastive loss to obtain more discriminative modality representations.**

obtain the cross-modal fusion representation  $\mathcal{H}_{gt}^1$ , as follows:

$$\begin{aligned} h_{gt}^1 &= \text{Concat}[H_{G'}^{g,1}; H_{L'}^{t,1}], \\ \mathbf{H}_{gt}^1 &= \text{Concat}(\text{avg}(\sum_{k=1}^K (\sigma(\omega_k \otimes h_{gt}^1) \otimes h_{ht}^1))), \\ \mathcal{H}_{gt}^1 &= \mathbf{H}_{gt}^1 \oplus \sigma(h_{G'}^{g,1}), \end{aligned} \quad (11)$$

where  $\sigma$  is an activation function,  $K$  is the number of heads,  $\oplus$  is element-wise addition, and  $\odot$  is multiplication operation.

Following above steps, taking the global textual feature  $h_G^{g,2}$  and local visual features  $h_L^{t,2}$  of  $e_2$  as inputs, we can obtain the another cross-modal fusion representation  $\mathcal{H}_{gt}^2$ . The graph-to-textual alignment score is then computed as:

$$S_{gt} = f(\mathcal{H}_{gt}^1, \mathcal{H}_{gt}^2). \quad (12)$$

In the same way, we can obtain the other cross-modal alignment scores  $S_{tg}, S_{tv}, S_{vt}$ . Finally, we combine the above score to get the overall cross-modal alignment result  $S_c$ .

$$S_c = (S_{gt} + S_{tg} + S_{tv} + S_{vt})/4. \quad (13)$$

### 3.4 Training

After obtaining the multi-grained decision scores, we employ a contrastive loss function for joint training. The goal is to ensure that positive samples receive higher alignment scores, while negative samples are assigned lower scores. In our setting, the pre-aligned

seed set serves as positive samples, while other non-aligned pairs are treated as negatives. Formally, for the  $i$ -th entity  $e_i^1 \in E_1$  in a mini-batch  $\mathcal{B}$ , the positive set is defined as  $\mathcal{P}_2^i = \{e_i^2 | e_i^2 \in E_2\}$ , where  $(e_i^1, e_i^2)$  is an aligned pair,  $E_1$  and  $E_2$  denote the sets of entities in MMKGs  $G_1$  and  $G_2$ , respectively. The negative set includes two parts, intra-graph negatives from the graph  $G_1$ :  $\mathcal{N}_j^1 = \{e_j^1 | \forall e_j^1 \in E_1, j \neq i\}$ , and cross-graph negatives from the graph  $G_2$ :  $\mathcal{N}_j^2 = \{e_j^2 | \forall e_j^2 \in E_2, j \neq i\}$ . Then the loss is defined as follows:

$$\mathcal{L}_u = \frac{\exp(S(e_i^1, e_i^2))}{\exp(S(e_i^1, e_i^2)) + \sum_{e' \in \{\mathcal{N}_j^1, \mathcal{N}_j^2\}} \exp(S(e_i^1, e'))}, \quad (14)$$

where  $\mathcal{M}(\cdot)$  represents the total alignment score. However, relying solely on the total score may cause one alignment module to dominate, leading the model to depend excessively on that score in order to achieve faster fitting. To this end, we assign an independent alignment loss to each alignment score, defined as follows:

$$\mathcal{L}_{\mathcal{A}} = \frac{\exp(\mathcal{S}_{\mathcal{A}}(e_i^1, e_i^2))}{\exp(\mathcal{S}_{\mathcal{A}}(e_i^1, e_i^2)) + \sum_{e' \in \{\mathcal{N}_j^1, \mathcal{N}_j^2\}} \exp(\mathcal{S}_{\mathcal{A}}(e_i^1, e'))}, \quad (15)$$

where  $\mathcal{A} \in \{g, t, v, gt, tv, vt\}$ .

### 3.5 Intra-modal Contrastive Learning

In addition, to obtain more discriminative embeddings that can accurately capture intra-modal differences, we introduce an intra-modal contrastive loss [12]. This loss pulls closer the representations of aligned seed pairs within the same modality while pushing apart those of unaligned pairs, thereby enhancing the reliability of alignment scores from different modules. We take the graph structural modality as an example. For positive pair  $(e_i^1, e_i^2)$ , we define the intra-modal contrastive loss as follows:

$$q_g(e_i^1, e_i^2) = \frac{\delta(e_i^1, e_i^2)}{\delta(e_i^1, e_i^2) + \sum_{e' \in \{N_j^1, N_j^2\}} \delta(e_i^1, e')}, \quad (16)$$

where  $\delta(u, v) = \exp(u^T \cdot v / \mathcal{T})$ , and  $\mathcal{T}$  is a temperature coefficient. Notably, the distribution is directional and asymmetric. Therefore, the loss for another direction is defined as  $q_t(e_2^i, e_1^i)$  analogously. The overall contrastive loss is then defined as:

$$\mathcal{L}_g^{cl} = -\mathbb{E}_{i \in \mathcal{B}} \log \left[ \frac{1}{2} (q_g(e_1^i, e_2^i) + q_g(e_2^i, e_1^i)) \right]. \quad (17)$$

Similarly, for the textual and visual modalities, we can define  $\mathcal{L}_g^{CL}$  and  $\mathcal{L}_v^{CL}$  respectively, and the final intra-modal contrastive loss is the average of all modalities, denoted as:

$$\mathcal{L}^{cl} = \sum_{m \in \{g, t, v\}} \mathcal{L}_m^{cl}. \quad (18)$$

### 3.6 Optimization Objective

The overall loss of the *DMEA* is formulated as:

$$\mathcal{L} = \mathcal{L}_u + \sum_{\mathcal{A} \in \mathcal{S}} \mathcal{L}_{\mathcal{A}} + \mathcal{L}^{cl}, \quad (19)$$

where  $\mathcal{S} = \{S_g, S_t, S_v, S_{gt}, S_{tg}, S_{tv}, S_{vt}\}$  denotes the set of all alignment scores.

## 4 Experiments Evaluation

### 4.1 Experimental Settings

**4.1.1 Datasets.** In the experiments, we selected two widely used cross-KG datasets, FB15K-DB15K and FB15K-YAGO15K, to evaluate the effectiveness of our proposed framework. FB15K-DB15K is constructed by aligning a subset of Freebase (FB15K) with DBpedia (DB15K), where entities are enriched with multi-modal information such as textual descriptions and images. FB15K-YAGO15K is built in a similar manner by linking entities between Freebase (FB15K) and YAGO (YAGO15K). We allocate 20% of the aligned entity pairs as training data, with the remaining data used for testing. Please refer to the appendix A.1 for detailed data statistics.

**4.1.2 Baselines.** We compared our method with various both representative and state-of-the-art approaches. Specifically, for traditional KGs, we select four representative methods: GCN-Align [31], MixTEA [34], SEA [22], RAC [39] and BootEA [25]. For MMKGs, we select eight representative methods: MMEA [2], and EVA [19], MCLEA [18], MEAFormer [6], PCMEA [29], SGMEA [7] and RICEA [16]. In addition, we rerun the official implementation of PCMEA and report the updated results. The results of the remaining baselines are directly taken from their original papers.

**4.1.3 Evaluation Metrics.** We perform alignment evaluation by computing alignment scores between entity pair. These scores are then used to calculate standard metrics, including Hits@k (H@k), Mean Reciprocal Rank (MRR). H@k measures the probability that the ground-truth entity appears within the top-k ranked candidates and we set  $k = \{1, 5, 10\}$  in this paper. MRR reflects the average reciprocal rank of the ground-truth entity. Therefore, higher values of H@k and MRR indicate better performance.

**4.1.4 Implementation Details.** We conduct all experiments on two 32G Tesla V100 GPUs, and use the AdamW optimizer within the deep learning framework PyTorch. Our model weights are initialized with pre-trained CLIP-Vit-Base-Patch32 and two-layer graph attention network. For graph structural modality, we set the dimensionality of node representations to 300. For textual modality, we restrict the maximum input length to 40 tokens and set the dimensionality of the textual feature representation to 512. For visual modality, all images are resized to a resolution of 224×224, and the dimensionality of the visual feature representation is set to 96. The parameters in the alignment layer are set to 96 for all two datasets.

### 4.2 Main Results

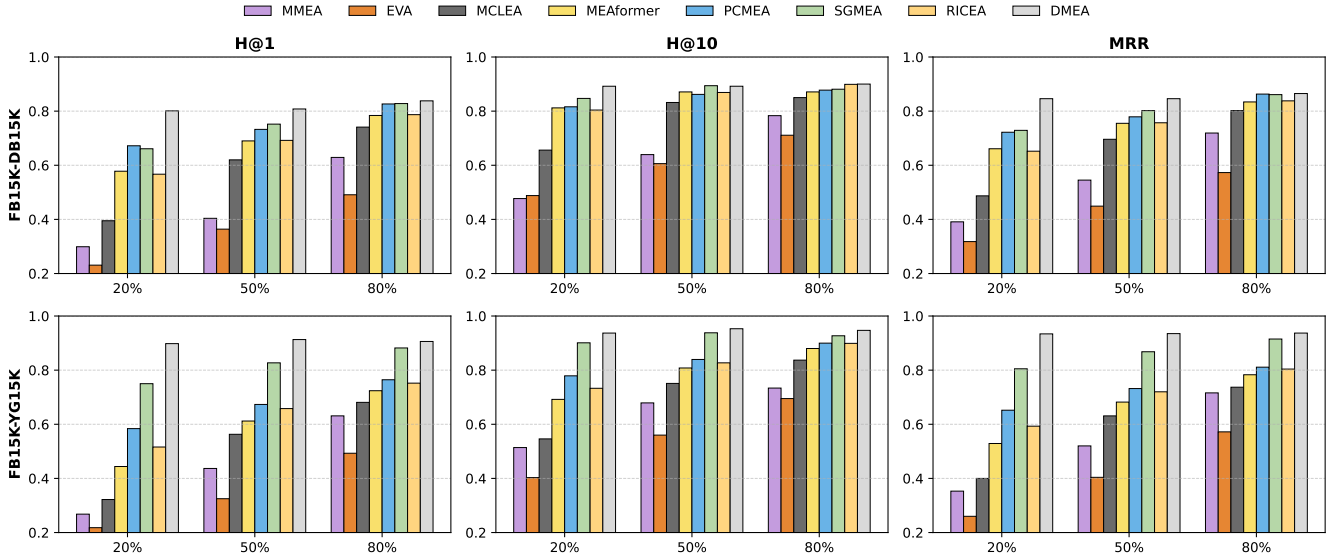
We compare *DMEA* with a series of baseline methods on two benchmark datasets. Table 1 reports the performance. Overall, *DMEA* achieves the best performance among all models, demonstrating its superiority in addressing the MMEA task. Based on the results presented in Table 1, we make the following key observations and analyses: (1) First, compared with traditional EA methods, multi-modal approaches show a clear advantage. All multi-modal methods following MEAformer outperform the SOTA traditional model, BootEA, validating the effectiveness of leveraging the rich and diverse information provided by multi-modal knowledge graphs for alignment. (2) Second, compared with existing MMEA baselines, *DMEA* exhibits highly competitive performance. While previous approaches based on feature-level fusion—such as PCMEA, SGMEA, and RICEA—have achieved notable improvements by incorporating high-quality pseudo labels, structure-guided representations, or modality-wise dynamic weighting, *DMEA* further surpasses them, improving Hit@1, Hit@10, and MRR by at least 13%, 5%, and 12%, respectively. (3) Finally, our proposed *DMEA* achieves outstanding Hits@1 scores of 0.804 and 0.898 on the two datasets using only 20% of aligned entity pairs for training. Such substantial improvements clearly demonstrate the potential of decision-level fusion in advancing the MMEA task.

### 4.3 Ablation Study

To assess the contribution of each component in *DMEA*, we conducted two groups of ablation experiments. In the first group, we remove the alignment scores of different modules, while in the second group, we eliminate individual loss terms. The results are presented in Table 2, from which several key observations can be drawn: (1) Every module plays a vital role in maintaining high performance, where textual alignment score is the most influential, highlighting the essential role of semantic description. (2) Removing any single alignment score results in a clear degradation, confirming that balanced decision fusion across modalities is critical for robust entity alignment. (3) Removing any single alignment module

**Table 1: We compare DMEA against traditional EA and most advanced MMEA approaches on the two benchmark datasets using 20% aligned entity pairs. The best results are highlighted in bold and the second best results are underlined. The “-” denotes that the results are not available.**

| Seeds | Model     | FB15K-DB15K  |              |              |              | FB15K-YAGO15K |              |              |              |
|-------|-----------|--------------|--------------|--------------|--------------|---------------|--------------|--------------|--------------|
|       |           | Hits@1       | Hits@5       | Hits@10      | MRR          | Hits@1        | Hits@5       | Hits@10      | MRR          |
| 20%   | GCN-Align | 0.043        | 0.110        | 0.155        | 0.082        | 0.023         | 0.072        | 0.107        | 0.053        |
|       | MixTEA    | 0.115        | 0.233        | 0.299        | 0.178        | 0.085         | 0.177        | 0.234        | 0.136        |
|       | SEA       | 0.170        | 0.335        | 0.425        | 0.255        | 0.141         | 0.287        | 0.371        | 0.218        |
|       | RAC       | 0.203        | 0.36         | 0.432        | 0.281        | 0.151         | 0.281        | 0.345        | 0.216        |
|       | BootEA    | 0.323        | 0.499        | 0.579        | 0.41         | 0.234         | 0.374        | 0.445        | 0.307        |
|       | MMEA      | 0.265        | 0.451        | 0.541        | 0.357        | 0.234         | 0.398        | 0.480        | 0.317        |
|       | EVA       | 0.231        | 0.341        | 0.488        | 0.318        | 0.218         | 0.302        | 0.403        | 0.26         |
|       | MCLEA     | 0.395        | 0.595        | 0.656        | 0.487        | 0.322         | 0.495        | 0.546        | 0.4          |
|       | MEAformer | 0.578        | -            | 0.812        | 0.661        | 0.444         | -            | 0.692        | 0.529        |
|       | PCMEA     | 0.672        | <u>0.781</u> | 0.816        | 0.722        | 0.584         | <u>0.733</u> | 0.779        | 0.652        |
|       | SGMEA     | 0.661        | -            | <u>0.847</u> | <u>0.729</u> | <u>0.750</u>  | -            | <u>0.901</u> | <u>0.805</u> |
|       | RICEA     | 0.567        | -            | 0.804        | 0.652        | 0.516         | -            | 0.733        | 0.593        |
|       | DMEA      | <b>0.804</b> | <b>0.880</b> | <b>0.896</b> | <b>0.849</b> | <b>0.898</b>  | <b>0.934</b> | <b>0.957</b> | <b>0.934</b> |



**Figure 4: Label dependency analysis results: comparison of different multi-modal methods on FB15K-DB15K and FB15K-YAGO15K datasets under 20%, 50%, and 80% alignment seed (label) settings.**

loss also leads to a noticeable performance drop, indicating that preventing one modality from dominating the others is a reasonable and effective strategy. (4) Excluding the intra-modal contrastive loss slightly decreases performance, demonstrating the importance of maintaining high discriminability within modality-specific representations.

#### 4.4 Label Dependency Analysis

Since annotating aligned entity pairs requires substantial human effort, evaluating a model’s dependence on the ratio of aligned pairs is an important aspect. Hence, we train the model using 20%, 50%, and

80% of the aligned entity pairs, respectively. In addition, to ensure a fair evaluation, we conduct experimental comparisons exclusively on MMEA-based methods. Figure 4 presents the comparative results of all MMEA methods under these settings. Overall, DMEA consistently demonstrates clear performance superiority across different proportions of training data. Moreover, even when the amount of training data decreases, DMEA maintains stable performance without significant degradation—contrasting sharply with most existing methods, whose performance drops notably under data-scarce conditions. For example, on the FB15K-DB15K dataset, when 50% of aligned entity pairs are used for training, SGMEA

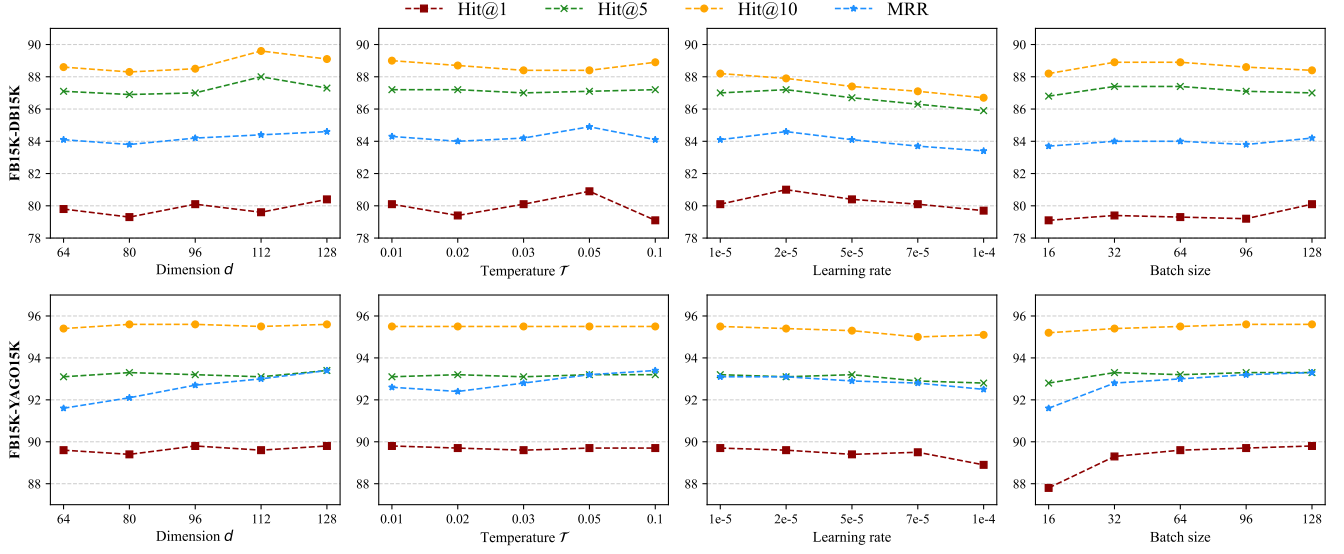


Figure 5: Parameter sensitivity analysis on the embedding dimension  $d$ , the temperature parameter  $\mathcal{T}$ , learning rate  $lr$  and batch size under both FB15K-DB15K and FB15K-YAGO15K dataset.

Table 2: Ablation Study of the key components in DMEA on FB15K-DB15K dataset with 20% pre-aligned seed set.

| Method                          | FB15K-DB15K  |              |              |              |
|---------------------------------|--------------|--------------|--------------|--------------|
|                                 | Hits@1       | Hits@5       | Hits@10      | MRR          |
| <i>DMEA</i>                     | <b>0.804</b> | <b>0.880</b> | <b>0.896</b> | <b>0.849</b> |
| w/o $\mathcal{S}_g$             | 0.783        | 0.874        | 0.8391       | 0.768        |
| w/o $\mathcal{S}_t$             | 0.713        | 0.809        | 0.864        | 0.764        |
| w/o $\mathcal{S}_o$             | 0.776        | 0.856        | 0.819        | 0.756        |
| w/o $\mathcal{S}_{gt}$          | 0.781        | 0.875        | 0.832        | 0.769        |
| w/o $\mathcal{S}_{to}$          | 0.783        | 0.866        | 0.821        | 0.759        |
| w/o $\mathcal{L}_m^{cl}$        | 0.794        | 0.872        | 0.868        | 0.820        |
| w/o $\mathcal{L}_{\mathcal{A}}$ | 0.786        | 0.873        | 0.839        | 0.770        |

achieves a higher Hits@10 score than RICEA. However, when the training ratio increases to 80%, their performance ranking reverses, indicating that such methods are highly dependent on the quantity of aligned entities. These results highlight that our approach effectively exploits multimodal knowledge, enabling robust performance even with limited supervision.

#### 4.5 Parameter Sensitivity Analysis

We evaluate the sensitivity of *DMEA* with respect to several hyperparameters, including the projection dimension  $d$ , temperature  $\mathcal{T}$ , learning rate, and batch size. The experimental results are presented in Figure 5. (1) *Effect of projection dimension*. As shown in the first column, the model performance varies moderately with different projection dimensions. On two datasets, the performance slightly increases as  $d$  grows, which indicates that a moderate projection

dimension enables richer feature representation while avoiding redundancy that may hinder generalization. (2) *Effect of temperature  $\mathcal{T}$* . Then, as  $\mathcal{T}$  increases from 0.01 to 0.1, the overall performance reaches a peak at 0.05 on FB15K-DB15K datasets and remains steady on FB15K-YAGO15K datasets, which shows that a reasonable temperature coefficient is crucial for performance. (3) *Effect of learning rate*. The performance trend under different learning rates exhibits a clear peak at  $2 \times 10^{-5}$  on both datasets. These results demonstrate that *DMEA* benefits from a carefully tuned learning rate to balance adaptation and stability during fine-tuning and a smaller learning rate may slow convergence. (4) *Effect of batch size*. Finally, we assess the influence of batch size. On both datasets, a larger batch size consistently improves model performance, especially for Hit@1 and MRR. This is because *DMEA* adopts in-batch contrastive learning, and a larger batch size introduces more diverse negative samples within each iteration, thereby enhancing the discriminative power of the learned embeddings.

#### 5 Conclusion

In this paper, we analyze the limitations of existing feature-fusion methods and propose *DMEA*, the first decision-fusion-based framework for MMEA. *DMEA* explicitly models multiple alignment relationships between entity pairs by leveraging both global and local features, and then performs entity alignment through decision fusion. Experimental results on two benchmark datasets, FB15K-DB15K and FB15K-YAGO15K, demonstrate that *DMEA* achieves improvements of 13.2% and 14.8% over the existing SOTA models at hit@1 value, respectively, thereby validating the significant advantages and practical potential of decision fusion in MMEA.

#### Acknowledgments

This work is supported by National Science and Technology Major Project (2026ZD16011200).

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## A Appendix

| Entity Pair                                                                                                                                                                                                                                                                                                    | Topk_id                                                                                                                                                                                                                                                                                                        | $S_g(R)$   | $S_v(R)$  | $S_t(R)$  | $S_{gt}(R)$ | $S_{tg}(R)$ | $S_{tv}(R)$ | $S_{vt}(R)$ |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-----------|-------------|-------------|-------------|-------------|
| Text:<br>Cormac McCarthy's date of birth is 1933.<br><br>Graph:<br>{influenced_by William Faulkner;<br>location Knoxville, Tennessee;<br>influenced David Foster Wallace...}<br><br>Vision:                                   | Text:<br>Cormac McCarthy's birth Date is 1933-07-20.<br><br>Graph:<br>{birthPlace Rhode_Island; influencedBy William_Faulkner; influencedBy James_Joyce...}<br><br>Vision:                                                    | 1.2983(1)  | 2.2421(3) | 1.8224(3) | 2.1634(1)   | 2.0845(2)   | 3.0191(1)   | 1.0425(3)   |
| Text:<br>Joseph Conrad's death Date is 1924-08-03.<br>Joseph Conrad's birth Date is 1857-12-03.<br><br>Graph:<br>{birthPlace Russian_Empire; occupation Short_story; influenced William_Faulkner...}<br><br>Vision:           | Text:<br>Joseph Conrad's death Date is 1924-08-03.<br>Joseph Conrad's birth Date is 1857-12-03.<br><br>Graph:<br>{birthPlace Russian_Empire; occupation Short_story; influenced William_Faulkner...}<br><br>Vision:           | 0.2985(2)  | 1.5738(4) | 2.2443(1) | 1.4144(2)   | 2.1898(1)   | 1.0887(4)   | 0.7932(4)   |
| Text:<br>Cormac McCarthy's birth Date is 1933-07-20.<br><br>Graph:<br>{birthPlace Rhode_Island; influencedBy William_Faulkner; influencedBy James_Joyce...}<br><br>Vision:                                                    | Text:<br>Mervyn C. Cooper's birth Year is 1893.<br>Mervyn C. Cooper's death Date is 1973.<br><br>Graph:<br>{birthPlace Florida; occupation Film_producer; almaMater Newell_Academy...}<br><br>Vision:                         | -0.2057(3) | 2.6124(1) | 2.1904(2) | 0.0968(4)   | 0.4945(3)   | 2.3342(3)   | 1.5805(2)   |
| Text:<br>William Cameron Menzies's birth Date is 1896-<br>William Cameron Menzies's death Date is 1957-<br><br>Graph:<br>{birthPlace New_Haven; occupation Production_designer; education Yale_University...}<br><br>Vision:  | Text:<br>William Cameron Menzies's birth Date is 1896-<br>William Cameron Menzies's death Date is 1957-<br><br>Graph:<br>{birthPlace New_Haven; occupation Production_designer; education Yale_University...}<br><br>Vision:  | -0.4481(4) | 2.4106(2) | 1.6968(4) | 0.8732(3)   | -0.1500(4)  | 2.5288(2)   | 1.9933(1)   |

Figure 6: A case study for DMEA.

### A.1 Details of datasets

Table 3 shows the detailed statistics of the two datasets.

Table 3: Statistics of two MMEA datasets, with EA pairs representing the reference entity alignments.

| Dataset       | KG      | #Ent. | #Rel. | #Img  | #EA pairs |
|---------------|---------|-------|-------|-------|-----------|
| FB15K-DB15K   | FB15K   | 14951 | 1345  | 13444 | 12848     |
|               | DB15K   | 12842 | 279   | 12837 |           |
| FB15K-YAGO15K | FB15K   | 14951 | 1345  | 13444 | 11199     |
|               | YAGO15K | 15404 | 32    | 11194 |           |

### A.2 Case study

For a more illustrative demonstration of the proposed DMEA, we conduct a case study. As illustrated in Figure 6, the first column shows a pair of correctly aligned entities, while the second column lists the top four entities most similar to "Cormac McCarthy". The remaining seven columns present the actual alignment scores produced by each alignment module, where (R) indicates the ranking of entities based solely on that module's score. It can be observed that the textual and visual modules exhibit considerable variation when used independently; however, when combined with other modules, the model achieves higher discriminability. By further incorporating the structural modality, the model successfully identifies the correct alignment.